SARVAS — TECHNICAL DOSSIER & HACKATHON BRIEFING

Satellite SAR Oil Spill Detection & AIS Forensic Vessel Attribution Platform

Problem Statement: Smart India Hackathon SIH26143	**Stack:** React 19 + FastAPI + PyTorch + PostGIS / SQLite	**Generated:** 12 Oct 2026
Core Vision Model: U-Net (ResNet18) 95.89% Acc	**Kinematic Engine:** 4th-Order Runge-Kutta Drift + Anomaly Forest	**Statutory Reference:** Merchant Shipping Act 1958 Sec 356

1. Executive Summary & The Core Problem

Modern satellite Earth observation platforms can routinely detect dark radiometric anomalies on the sea surface. However, **the true operational challenge begins after detection**:

(1) False Alarms: Natural biogenic slicks, algal blooms, and low-wind calm sea surfaces create radar backscatter damping identical to petroleum hydrocarbons, wasting critical Coast Guard flight sorties.

(2) Hydrodynamic Drift Displacement: Marine oil slicks advect rapidly under surface currents and prevailing winds. By the time a satellite passes overhead (often 6 to 36 hours post-discharge), the slick has drifted kilometers away from the true discharge origin. *The vessel nearest to the slick in the satellite image is virtually never the culprit.*

(3) Stealth Discharges & Dark Ships: Rogue tanker operators deliberately switch off AIS transponders (dark gaps) during nocturnal bilge dumping or tank washing to escape legal attribution.

SARVAS solves this end-to-end by linking satellite radar segmentation with hydrodynamic backtracking, AIS trajectory reconstruction, dead reckoning across blackout gaps, and multi-factor explainable machine learning.

2. Technical Terms & Acronyms Glossary

Term / Acronym	Full Expansion	Technical Definition & Role in SARVAS
SAR	Synthetic Aperture Radar	Active microwave radar (Sentinel-1 C-band, 5.405 GHz). Penetrates clouds, darkness, and rain. Measures sea surface roughness via radar backscatter (sigma nought, σ° in dB). Oil damps capillary waves, appearing dark.
AIS	Automatic Identification System	VHF transponder system mandated by IMO SOLAS for ships >300 GT. Broadcasts dynamic kinematic telemetry: MMSI, SOG, COG, Heading, Lat/Lon, and navigation status every few seconds.
MMSI	Maritime Mobile Service Identity	Unique 9-digit vessel identifier transmitted over AIS, linked to IMO registry, flag state, vessel dimensions, and registered shipowner.
SOG	Speed Over Ground	Speed of vessel relative to sea floor (in knots). Critical for anomaly detection: tankers drop SOG to <2 knots while discharging oily bilge water.
COG	Course Over Ground	Direction of vessel motion relative to true North (0°–359°). Sharp heading changes (>45°) signal evasive routing or maneuvers during illicit dumping.
RK4	4th-Order Runge-Kutta Advection	High-precision numerical integration algorithm for solving ODEs. Simulates Lagrangian particle advection under time-varying ocean currents and wind fields with dt=600s.
EEZ	Exclusive Economic Zone	Sovereign maritime zone extending 200 nautical miles from coastline under UNCLOS Part V. Dictates Indian legal jurisdiction under Merchant Shipping Act 1958.
NOS-DCP	National Oil Spill Disaster Contingency Plan	Statutory operational framework administered by the Indian Coast Guard for multi-tier environmental disaster mobilization (Tier 1: <700t, Tier 2: 700-10,000t, Tier 3: >10,000t).
IoU	Intersection over Union	Evaluation metric for semantic segmentation: Area of Overlap / Area of Union. SARVAS achieves **0.7922 Mean IoU** across Oil, Look-alike, and Clean Sea classes.
Dead Reckoning	Kinematic Track Interpolation	Reconstruction of probable vessel trajectory across AIS transponder blackout gaps by projecting position forward using pre-gap and post-gap kinematic vectors.
Isolation Forest	Unsupervised Anomaly Model	Tree-based outlier detector that partitions spatial-temporal features (speed variance, sinuosity, acceleration) to detect anomalous vessel behavior without labeled illicit tracks.
KDE	Kernel Density Estimation	Non-parametric estimation of spatial particle cloud probability density, used to generate mathematical 50% and 90% confidence contours around spill origin.

3. The 9-Step Operational Science Chain

SARVAS does not rely on opaque heuristics. Every inference is rooted in peer-reviewed physics and reproducible algorithms:

- **Step 1: SAR Satellite Ingestion:** Calibrates Sentinel-1 Level-1 GRD SAR imagery, strips speckle noise via Lee filter, normalizes radar cross-section σ° values, and tiles into 256x256 inference patches.
- **Step 2: Deep U-Net Segmentation:** Executes multi-class PyTorch U-Net with ResNet18 encoder. Segments each pixel into **Class 0: Oil Slick**, **Class 1: Look-alike (biogenic)**, or **Class 2: Open Sea** (Validation accuracy: 95.89%).
- **Step 3: Slick Characterization & Geometry:** Extracts polygon boundaries, calculates surface area (km²), perimeter, elongation ratio, centroid coordinates, and estimates spill volume based on standard Bonn thickness matrices.
- **Step 4: Backward Lagrangian Drift Backtracking:** Injects 500 stochastic particles at slick centroid and integrates backward in time (-24h) via 4th-order Runge-Kutta advection driven by Copernicus CMEMS currents and ECMWF ERA5 10m surface winds.
- **Step 5: Origin Uncertainty Cone & 50%/90% Contours:** Computes Gaussian Kernel Density Estimation (KDE) over the back-tracked particle cloud to derive a mathematically honest origin cone, peak coordinates, and confidence contours.
- **Step 6: AIS Spatial-Temporal Intersection Filtering:** Queries all AIS vessel tracks within the origin bounding box and spill time window. Prunes straight-line cruising traffic and boundary-clipping ships, narrowing hundreds of vessels down to 2-5 candidates.
- **Step 7: Dead-Reckoning & Anomaly Forensics:** Reconstructs probable vessel tracks across intentional AIS transponder shutoff gaps. Evaluates speed drops (SOG), sharp turns (COG), loitering, and runs an Isolation Forest model.
- **Step 8: 6-Factor Explainable Suspect Attribution:** Computes transparent sub-scores across 6 physical factors (Proximity, Time Overlap, AIS Gap, Speed, Course, Route Deviation) to output a deterministic 0-100 suspect likelihood score.
- **Step 9: 48-Hour Forward Impact & Legal Evidence Dossier:** Simulates forward oil dispersion over India's 11,098.81 km coastline and coral reefs; compiles an official court-admissible 2-page PDF report with SHA-256 seal and statutory legal references.

4. Machine Learning & Model Training Deep-Dive

Model Architecture: Semantic segmentation U-Net utilizing a pre-trained resnet18 encoder backbone from segmentation-models-pytorch. The model operates on single-channel SAR backscatter intensity tiles (256×256 pixels).
Dataset Source: Trained on the authentic open-access **Zenodo Sentinel-1 SAR Oil Spill Dataset** (Krestenitis et al., Marine Pollution Bulletin), comprising Level-1 Ground Range Detected (GRD) C-band images from European Space Agency (ESA) acquisitions across global ocean basins.
Classes: 3-class segmentation — (0) Petroleum Oil Slick, (1) Natural Look-alike (biogenic films, low-wind damping), (2) Background Ocean Clutter.
Loss Function & Optimization: CrossEntropyLoss with Adam optimizer (learning rate 1e-4, weight decay 1e-5). Evaluated across per-class Intersection-over-Union (IoU) and pixel accuracy.

Metric	Epoch 1	Epoch 4	Epoch 8 (Best)	Operational Significance
Validation Accuracy	59.87%	95.83%	**95.89%**	Near-zero false alarm rate on background ocean.
Training Accuracy	40.56%	96.54%	**96.45%**	Robust convergence with no over-fitting.
Mean IoU (Overall)	0.2068	0.7219	**0.7922**	High overlap across all 3 classes.
Oil Class IoU	0.0219	0.2000	**0.4107**	Strong spatial boundary agreement on slicks.
Validation Loss	1.1572	0.4092	**0.3230**	Smooth cross-entropy descent.
Traced Model Memory	-	-	**0.38 MB RAM**	TorchScript traced model (unet_traced.pt) for cloud zero-latency.

5. Hydrodynamic Lagrangian Drift Engine (The Math)

Governing Equations: The total advection velocity vector acting on an oil particle is formulated as:
v_total = v_current + alpha * v_wind, where v_current is the sea surface velocity from Copernicus CMEMS, v_wind is the 10-meter atmospheric wind velocity from ECMWF ERA5, and alpha = 0.03 (the standard 3% empirical windage factor).
Numerical Integration (RK4): Rather than naive Euler stepping, SARVAS solves the advective trajectory using 4th-order Runge-Kutta: **x_(n+1) = x_n + (dt / 6) * (k1 + 2*k2 + 2*k3 + k4)** with time step dt = 600 s (10 minutes).
Stochastic Turbulent Diffusion: Lateral spreading is modeled via a random walk displacement: sigma = sqrt(2 * K * dt), where K = 50.0 m^2/s is the horizontal eddy diffusion coefficient.
Coastline Collision Boundary: Each particle step is intersected against India's sovereign high-resolution coastline LineStrings via Shapely prepared geometry; beached particles cease advecting.

6. Explainable Suspect Attribution Algorithm (6 Factors)

In compliance with maritime legal standards, SARVAS never labels a vessel as 'guilty'. Instead, it generates a transparent **Attribution Likelihood Score (0–100)** decomposed into 6 distinct physical factors. This provides Coast Guard boarding parties with clear evidentiary backing:

Factor Name	Weight	Measurement Formula / Threshold	Evidentiary Meaning
Proximity	**25%**	100 * (1 - dist_nm / 50); dist <= 1nm => 100	Haversine distance from vessel track to drift origin cone centroid.
Time Overlap	**20%**	100 * (hours_in_cone / window_hours)	Temporal intersection of vessel position with calculated spill time window.
AIS Transponder Gap	**20%**	min(gap_duration_min / 2, 100) if near spill	Detects intentional shutoffs (>60 min blackout near origin is heavily penalized).
Speed Anomaly	**15%**	min(events * 30 + (5 - min_sog) * 10, 100)	Detects speed drops below 2.0 knots consistent with bilge water pumping.
Course Anomaly	**10%**	min(events * 25 + max_turn_deg / 3, 100)	Sharp heading shifts (>45°) suggesting evasive maneuvers during dumping.
Route Deviation	**10%**	min(dev_nm * 5 + (sinuosity - 1) * 50, 100)	Unusual detours away from standard commercial shipping lanes.

7. Data Provenance: Real vs. Synthetic Breakdown

For hackathon evaluators and technical judges, complete transparency regarding data provenance is essential. Here is the exact truth table of what data in SARVAS is 100% authentic vs. what is synthetically modeled:

Asset / Subsystem	Type	Source & Provenance	Details & File Path
Sovereign EEZ Boundaries	**REAL**	Flanders Marine Institute / Marineregions.org	Authentic 200 nm Indian EEZ polygon (india_eez.geojson, 551 KB).
National Coastline	**REAL**	Natural Earth 10m Vector Dataset	High-res 11,098.81 km Indian mainland & islands (india_coastline.geojson, 786 KB).
Coral Reef Systems	**REAL**	UNEP-WCMC & Allen Coral Atlas	Global protected coral polyline & polygon baseline (coral_reefs.geojson, 12.3 MB).
ERA5 Surface Wind Vectors	**REAL**	ECMWF Copernicus Climate Data Store	Authentic 10m u/v wind field netCDF (era5_wind_india.nc, 584 KB).
Sentinel-1 SAR Test Scenes	**REAL**	European Space Agency (ESA) Copernicus Hub	6 calibrated C-band scenes + ground-truth masks (data/sar/demo_for_judges/).
U-Net Training Weights	**REAL**	Zenodo Open Oil Spill Repository	Trained on genuine Sentinel-1 SAR tiles (unet_best.pth, 54.7 MB).
Live AIS Vessel Stream	**REAL**	AISStream.io WebSocket Feed	Live global commercial vessel positions via API token in stream_ais.py.
Demo Dark Ship Scenarios	**SYNTHETIC**	Procedural Kinematic Seed Generator	Seeded demo ships with deliberate transponder shutoffs and speed drops for instant demo.
Offshore Current Grid Fallback	**SYNTHETIC**	Empirical Regional Baseline	0.25 m/s current vector (215° bearing) fallback when CMEMS 153MB NetCDF is omitted.

8. External APIs, AI Integration & Keys

SARVAS integrates three major cloud and data service APIs:

API / Service	Provider & Endpoint	Configuration Key	Operational Role in System
Qualcomm Cloud AI Playground	Cirrascale Enterprise Cloud https://aisuite.cirrascale.com	QUALCOMM_API_KEY QUALCOMM_MODEL=Llama-3.3-70B	Generates plain-language forensic attribution summaries for law enforcement. Uses strict system prompt guardrails: no hallucinated laws, pure factual explanations based on the 6 scoring factors.
AISStream.io	AISStream Global Network wss://stream.aisstream.io/v0	AISSTREAM_API_KEY	Live streaming WebSocket client that ingests real-time ship transponder messages (MMSI, position, speed, course) across the Indian Ocean in background workers.
Copernicus Climate Data Store (CDS)	ECMWF Copernicus CDS API https://cds.climate.copernicus.eu	CDS_API_KEY	Automated retrieval of ERA5 atmospheric reanalysis NetCDF grids (10-meter U and V wind components) to drive Lagrangian oil advection.

9. Statutory Legal Framework & Court Evidence Dossier

A key differentiator of SARVAS is its legal grounding. Instead of stopping at a software dashboard, SARVAS exports an **Official Maritime Evidence Dossier** formatted to statutory Indian Coast Guard and NTRO standards:

- **Merchant Shipping Act 1958 (Part XI-A, Section 356):** Governs containment of pollution of the sea by oil. Authorizes the Central Government and Coast Guard to direct vessels to prevent escape of oil and issue detention orders.
- **UNCLOS Part XII (Articles 194, 211, 220):** Establishes sovereign enforcement jurisdiction within the 200 nm Exclusive Economic Zone for marine pollution violations.
- **Cryptographic SHA-256 Evidence Seal:** Every exported PDF embeds a tamper-evident SHA-256 digital hash computed over the slick geometry, drift parameters, and suspect vessel records. Any post-export tampering invalidates the integrity digest.
- **Investigative Lead Safeguard:** Explicit legal disclaimer stamped on every dossier: all scores reflect mathematical circumstantial likelihood, requiring physical Coast Guard boarding party verification and oil sampling.

10. Role-Based Access Control (RBAC) & Operational Consoles

SARVAS provisions 4 tailored operational consoles built in React 19 + MapLibre GL:

Role / Authority	Account	Route	Operational Capabilities & Workflows
Tactical Interceptor (Indian Coast Guard)	coast_guard demo123	/dashboard/coastguard	Tactical radar viewer, dark spot classification, live vessel track replay, transponder blackout alerts, suspect ranking, and court PDF dossier export.
Regional Director (Environmental Center)	regional_mgr demo123	/dashboard/regional	48-hour forward dispersion simulation, coral reef vulnerability overlays, containment boom staging, and commercial fisheries loss breakdown.
National Executive (Maritime Authority)	authority demo123	/dashboard/authority	Multi-state coastal risk breakdown (Gujarat, Maharashtra, Goa, etc.), inter-agency communication, and high-level incident clearance.
Citizen / Public (Coastal Advisory)	public_user demo123	/	Open transparency portal: beach advisories, verified cleanup statuses, safety guidelines, and citizen spill reporting.

Key Hackathon Pitch Takeaway: *SARVAS is not just another segmentation viewer. It bridges the complete gap between spaceborne radar pixels and courtroom prosecution: Satellite Ingestion -> Deep U-Net -> Lagrangian RK4 Backtracking -> AIS Anomaly Mining -> 6-Factor Explainable Attribution -> Qualcomm Llama-3.3 Brief -> Court-Admissible Tamper-Proof Evidence Dossier.*